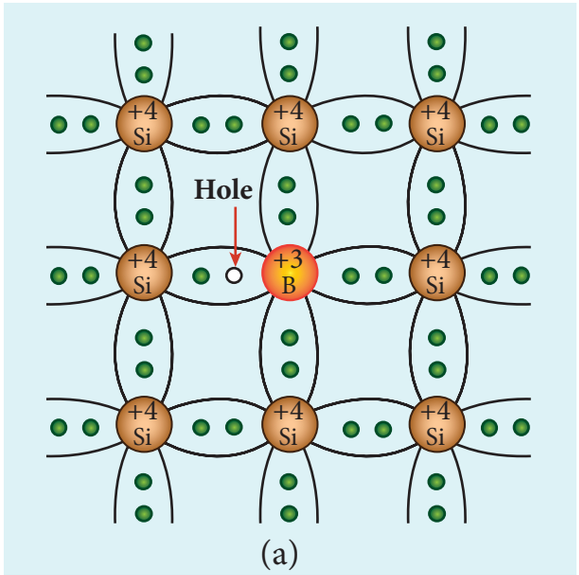


Figure 10.7 *p*-type extrinsic semiconductor
(a) Hole generated by the dopant
(b) Representation of acceptor energy level

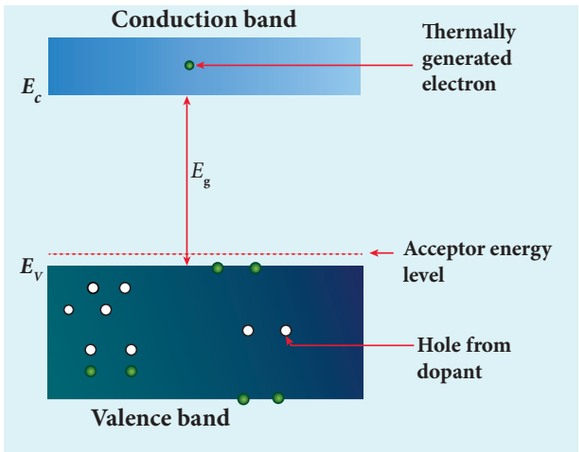


Figure 10.8 Thermally generated electron in the conduction band and the holes generated by the dopants in the valence band (*p*-type semiconductor)



The *n*-type and *p*-type semiconductors are neutral because only neutral atoms are doped to the intrinsic semiconductors.

10.3 DIODES

10.3.1 P-N Junction formation

i) Formation of depletion layer

A single piece of semiconductor crystal is suitably doped such that its one side is *p*-type semiconductor and the other side is *n*-type semiconductor. The contact surface between the two sides is called *p-n* junction. Whenever *p-n* junction is formed, some of the free electrons diffuse from the *n*-side to the *p*-side while the holes from the *p*-side to the *n*-side. The diffusion of charge carriers happens due to the fact that the *n*-side has higher electron concentration and the *p*-side has higher hole concentration. **The diffusion of the majority charge carriers across the junction gives rise to an electric current, called diffusion current.**

When an electron leaves the *n*-side, a pentavalent atom in the *n*-side becomes a positive ion. The free electron migrating into *p*-side recombines with a hole present in a trivalent atom near the junction and the trivalent atom becomes a negative ion. Since such ions are bonded to the neighbouring atoms in the crystal lattice, they are unable to move. As the diffusion process continues, a layer of positive ions and a layer of negative ions are created on either side of the junction accordingly. **The thin region near the junction which is free from charge carriers (free electrons and holes) is called depletion region** (Figure 10.9).

An electric field is set up between the positively charged layer in the *n*-side and

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